

Payroll and Depreciation

BUS123 Math M05 L01 pre-reading - Harborside Medical Center

Before class, read this short briefing so the calculations in the live activity feel like business decisions instead of isolated formulas. Harborside Medical Center is deciding how to manage weekly staffing costs while planning for long-lived medical equipment.

Why this matters

Managers need a clean view of two different cost rhythms: payroll costs that repeat every pay period and asset costs that are spread across years. Payroll helps Harborside staff patient care safely. Depreciation helps Harborside recognize that equipment wears out, becomes outdated, or loses value over time.

Key payroll vocabulary

Term	Meaning for Harborside Medical	Basic formula
Hourly pay	Employees are paid for hours worked, often with overtime eligibility.	hours x rate
Overtime pay	Extra pay for hours beyond the regular threshold, commonly 40 hours per week.	OT hours x rate x 1.5
Salary	Fixed annual pay converted into weekly, biweekly, semimonthly, or monthly checks.	annual salary / periods
Commission	Pay based on a percentage of sales, referrals, or measurable revenue.	sales x commission rate
Gross pay	Total earned before deductions.	regular + OT + commission

Payroll cycles

The pay cycle changes the size and timing of a paycheck, not the annual salary. A \$52,000 salary can be paid as \$1,000 weekly, \$2,000 biweekly, about \$2,166.67 semimonthly, or about \$4,333.33 monthly.

Salary per period = annual salary / number of pay periods

Overtime example

Suppose a Harborside nurse earns \$44 per hour and works 43 hours in one week. The first 40 hours are paid at the regular rate. The 3 overtime hours are paid at time-and-a-half.

Regular pay = 40 x \$44 = \$1,760

Overtime pay = 3 x \$44 x 1.5 = \$198

Gross pay = \$1,958

Manager question: If many employees are working overtime every week, is Harborside solving a temporary staffing issue or hiding a permanent headcount problem?

Different payroll systems create different incentives

Pay system	Best fit	Risk to watch
Hourly	Patient-care shifts, front desk coverage, variable schedules.	Overtime can rise quickly.
Salary	Managers, billing leaders, ongoing responsibility.	Workload may become invisible.
Piece rate or per visit	Task-based work, contract coverage, repeatable services.	Volume may be rewarded over quality.
Commission	Outreach, referrals, optional service growth.	Revenue incentive may need guardrails.
Salary plus commission	Stable role with a growth target.	Formula must be transparent.

Depreciation vocabulary

Depreciation is an accounting estimate of the cost of a long-lived asset used up during a period. It is not the same as cash leaving the bank this year. It helps Harborside match the cost of equipment to the years that benefit from that equipment.

Asset cost: the purchase price plus costs needed to get the asset ready for use.

Residual value: expected value at the end of useful life.

Useful life: how long the company expects to use the asset.

Accumulated depreciation: total depreciation recorded so far.

Book value: cost minus accumulated depreciation.

Straight-line depreciation

Straight-line depreciation spreads the depreciable cost evenly across useful life. Managers like it because it is predictable, easy to explain, easy to audit, and useful when an asset provides steady service.

$$\text{Annual depreciation} = (\text{asset cost} - \text{residual value}) / \text{useful life}$$

Harborside example

Harborside buys a diagnostic ultrasound unit for \$84,000. The expected residual value is \$12,000 after 6 years.

$$(\$84,000 - \$12,000) / 6 = \$12,000 \text{ annual depreciation}$$

Year	Cost	Depreciation expense	Accumulated depreciation	Book value
1	\$84,000	\$12,000	\$12,000	\$72,000
2	\$84,000	\$12,000	\$24,000	\$60,000
3	\$84,000	\$12,000	\$36,000	\$48,000
6	\$84,000	\$12,000	\$72,000	\$12,000

Prep questions for class

1. Which payroll type gives Harborside the most flexibility when patient volume changes week to week?
2. Why does overtime make total payroll rise faster than regular hours?
3. If an employee earns a fixed annual salary, why do weekly and monthly paychecks have different dollar amounts?
4. What is the difference between depreciation expense and accumulated depreciation?
5. Why might straight-line depreciation be a good fit for a medical device used steadily throughout the year?

Bring to class

Be ready to use the payroll management system to test a staffing choice. Your goal is not only to calculate the new payroll amount, but also to explain whether the change is operationally reasonable for a medical practice.